

Sumitomo Superconducting Compact Proton Cyclotron SC230

Sumitomo Heavy Industries, Ltd.

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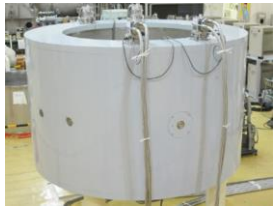


Accelerators



SHI Cryogenics Group

Cryocoolers



Superconducting magnets

- Accelerator manufacturing for 50 years

1972- Research accelerator

1981- PET cyclotron

1988- Injector for heavy ion radiotherapy

1996- Proton therapy system

2009- BNCT system

K140 AVF cyclotron（RCNP, Osaka Univ.）



Medical accelerators

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PET



Linac for heavy ion radiotherapy



BNCT



Radionuclide therapy



Proton therapy



Outline

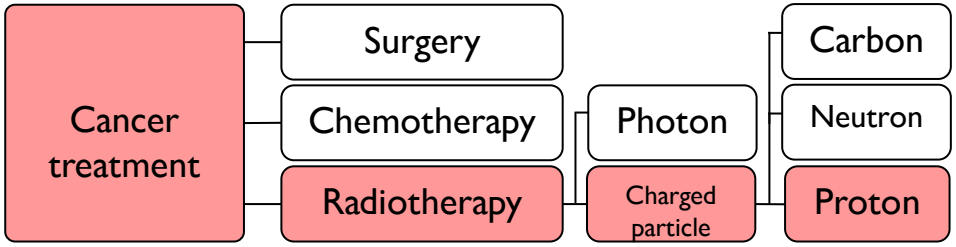
- **Introduction**
  - Proton therapy, PT system
- **SC230 : Superconducting AVF cyclotron**
  - Concepts of SC230
  - SC magnet
  - Mapping
  - Commissioning
  - SCIF (screening-current-induced field)
- **Conclusion**

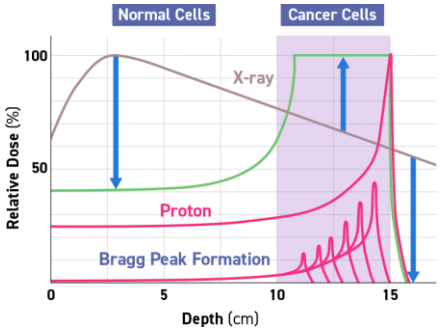
Proton therapy (PT)

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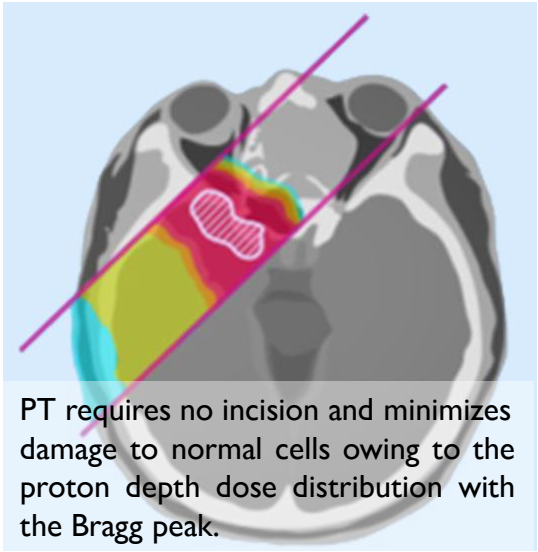
QOL；Quality Of Life

- PT is a cancer treatment that maintains the patient's QOL





Dose distribution with PT



PT requires no incision and minimizes damage to normal cells owing to the proton depth dose distribution with the Bragg peak.

PT facilities

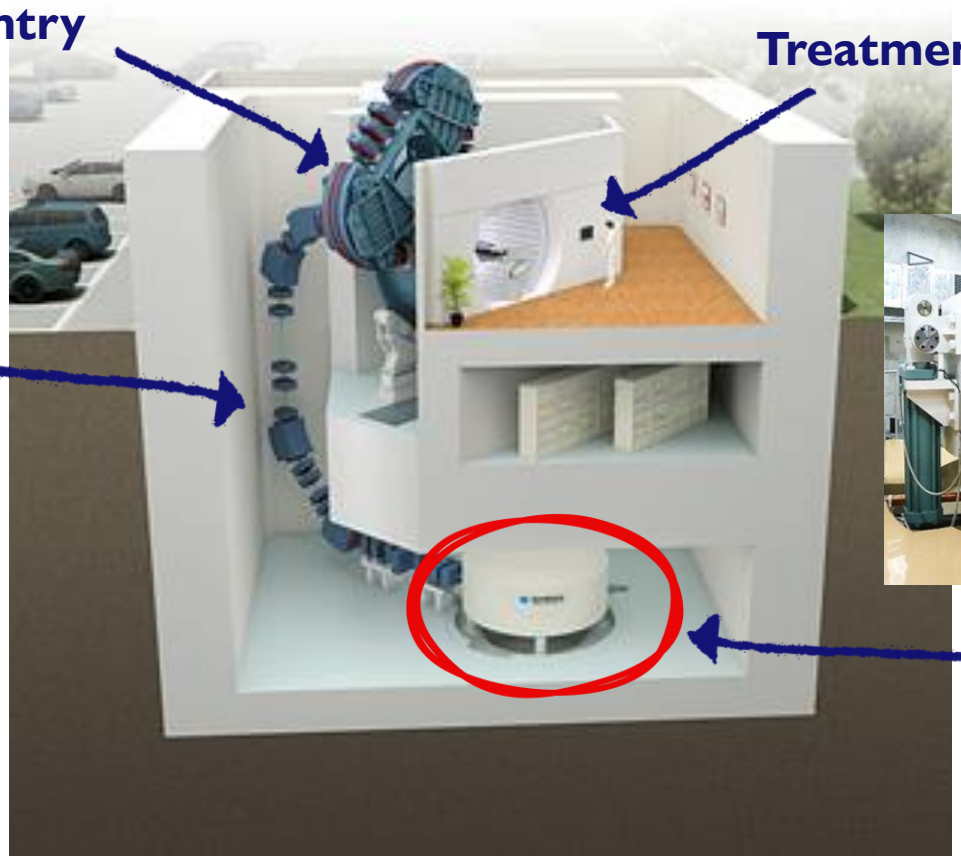


Recently, PT has gained prevalence in cancer treatment.

PT system

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SHI's PT system



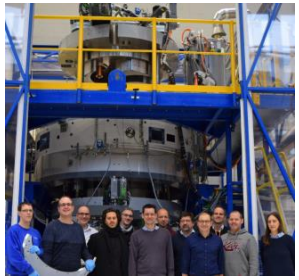
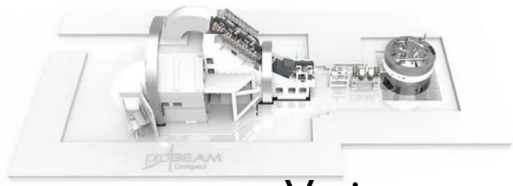


The large device size limits its use in hospitals.
Manufacturers have been miniaturized components in recent years.

Commercial SC cyclotrons for PT

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- Commercial superconducting(SC) cyclotrons for PT



Varian

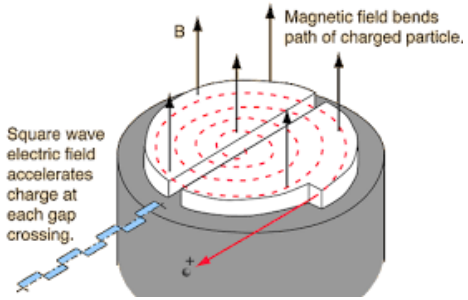


Mevion



IBA

- A high field by a SC magnet makes the accelerator compact.



$$\rho = \frac{mv}{qB}$$

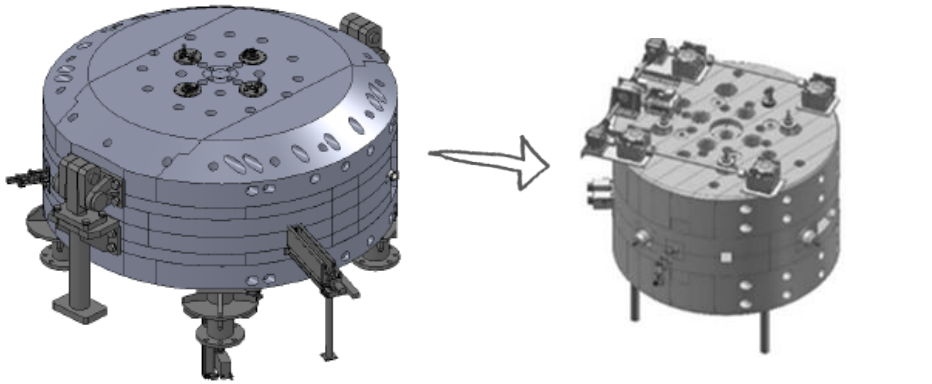
ρ : Radius
B : Magnetic field
m : Mass of particle
v : Velocity
q : Charge

Commercial SC accelerators for PT have successfully developed.

Our new cyclotron SC230

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- SC230 : Superconducting AVF cyclotron for PT



Conventional model (P235)

Developed model (SC230)

Parameters	P235	SC230	Ratio
Weight	220 t	65 t	× 0.3
Power consumption	330 kW	< 200 kW	× 0.6
Maximum beam current	300 nA	1000 nA	× 3.3

SC230 has the characteristics of
“Compact size”, “Energy saving”, “High beam current"

Design of SC230

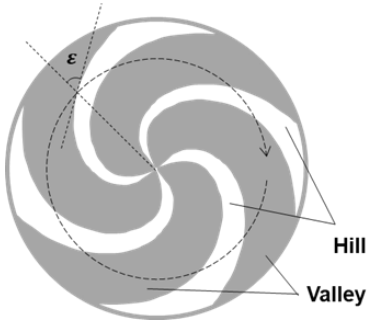
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H. Tsutsui, et al. (2013)

Parameters	Specifications
Type	AVF cyclotron
Particle species	Proton
Beam energy	233 - 238 MeV (Fixed)
Maximum beam current	> 1000 nA
Beam structure	CW
Extraction radius	0.6 m
Yoke size	Diameter: 2.8m Hight: 1.7 m
Yoke weight	65 t
RF frequency	95.3 MHz (h = 2)
Dee voltage	50 kV (Inside) 75 kV (Outside)
Number of dees	2
RF power	< 70 kW
Total power consumption	< 200 kW



- 4 sectors
  with large spiral angles of max. ~70 degrees.
  with small hill gap of min. ±6mm



$$\nu_z^2 \cong -\frac{r}{B_z}\left(\frac{\partial r}{\partial B_z}\right) + F^2(1+2\tan^2\varepsilon), \qquad F^2 = \frac{(B_{\mathrm{h}}-\bar{B})(\bar{B}-B_{\mathrm{v}})}{\bar{B}^2}$$

Aiming to realize these concepts,
the design and fabrication of the components proceeded.

Design of SC230

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- RF cavities
  with only two dee
  with low dee voltage of 50-75 kV
- Beam extraction
  using the precessional extraction method (using resonance of $\nu_r = 1$)

Aiming to realize these concepts,
the design and fabrication of the components proceeded.

Fabricating of components

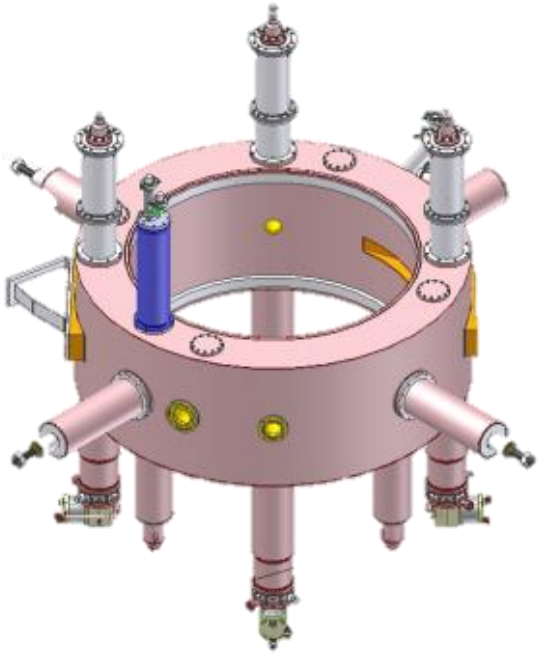
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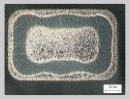
J. Yoshida, et al. (2019)

- **Cryogen-free NbTi superconducting magnet**

  using conduction cooling method by four 4K-GM cryocoolers





Parameters		Specifications
SC wire 	Material	NbTi/OFC
	Thickness	2.10 mm
	Width	3.40 mm
Coil	Configuration	Two circular coils with axial gap
	Number of turns	2208 Turns/Coil
Coil cooling	method	Conduction cooling
	cryocooler	Four 4K-GM cryocoolers
Coil current		~ 440 A (Max. 488 A)
Coil temperature		4 K
Cooling time from RT		14 days
Stored energy		5.3 MJ
Recovery time from quench		17 hours

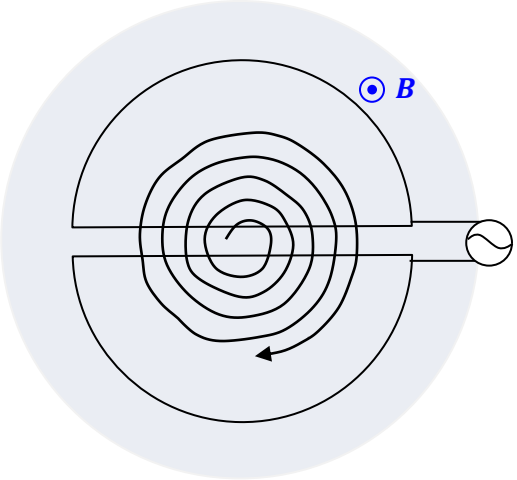
A SC magnet with high safety and maintainability has been developed.

Isochronous magnetic field

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- **Isochronism in classical cyclotrons**

$$qvB = mv\omega \quad \text{(Lorentz force) (centripetal force)} \qquad \therefore \omega = \frac{qB}{m}$$



- **Influence of relativistic effects**

$$\omega = \frac{qB}{m_0\gamma}$$

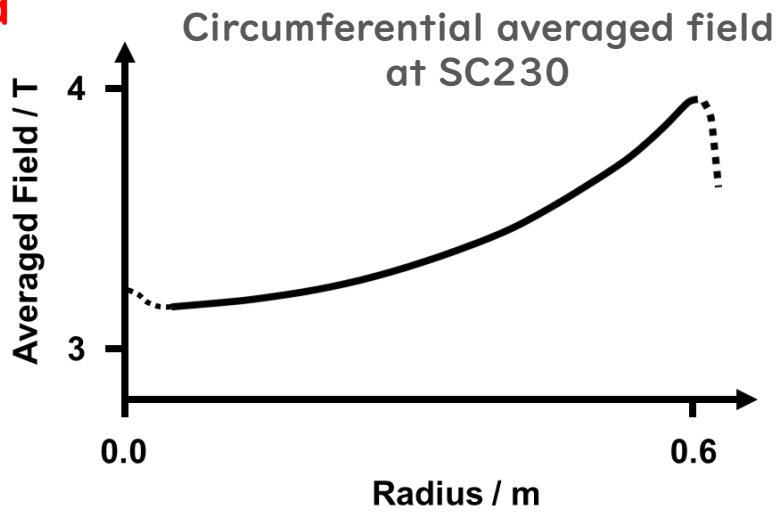
$$\gamma = \frac{1}{\sqrt{1-\beta^2}}, \qquad \beta = v/c$$

γ : Lorentz factor
c : speed of light

- **isochronous magnetic field**

$$B = \gamma B_0$$

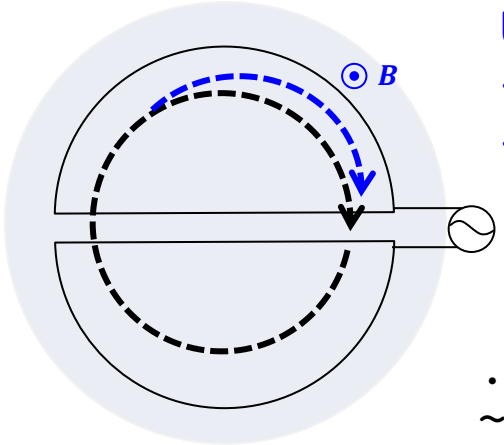
$$\omega = \frac{q(\gamma B_0)}{m_0\gamma} = \frac{qB_0}{m_0}$$



Mapping and field adjustment

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- Disturbance of isochrony

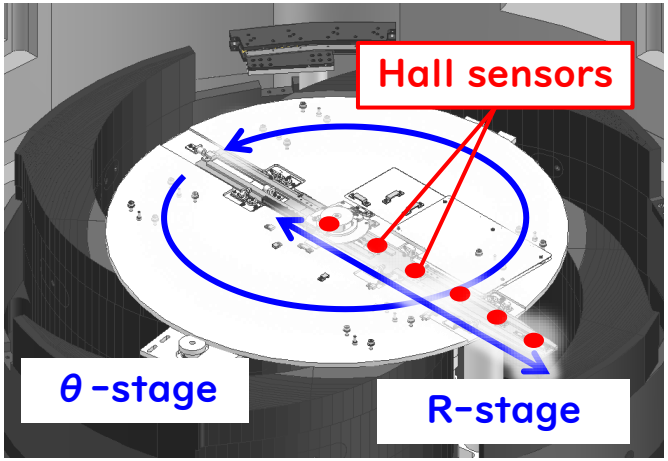


Uncertainty of magnetic field
→ Orbit deviation
→ Phase shift

$$\sin(\delta\phi) \sim -\frac{\omega_{\mathrm{RF}}}{V_{\mathrm{acc}}} \int dS\delta B$$

- The allowable field uncertainty
~±2Gauss for Circumferential mean field

- Mapper for SC230

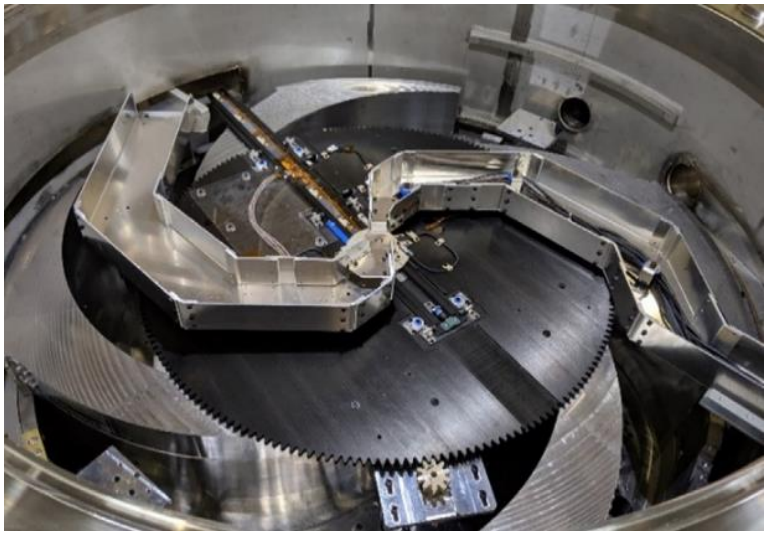


Mapping and field adjustment

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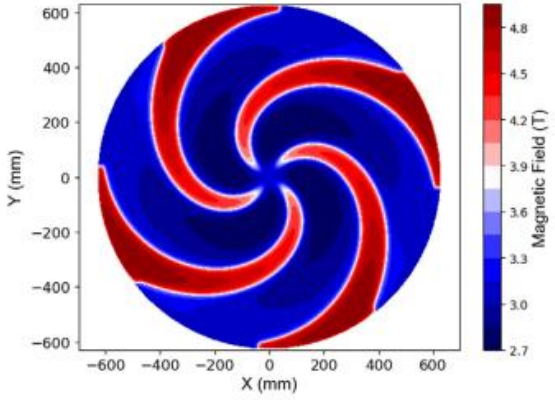
Y. Ebara, et al. (2020)

- **Mapping and field adjustment for SC230**

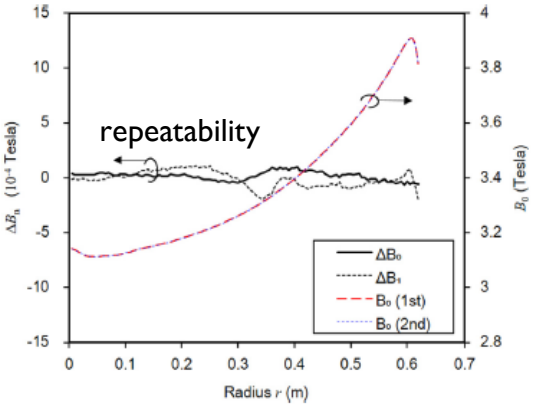


Parameters	Specifications
Sensor type	Hall probes
Number of sensors	6
Type of stages	Linier and rotate stages
Material of stage	CFRP
Actuators	USMs
Measurement points	~52 000
Measurement area	ϕ1,260 mm
Measurement time	2.5 hours
Adjustment methods	Pole machining Shimming Coil centering

Measured field



Isochronous field



Required magnetic field was formed by mapping and adjustment.

Commissioning

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Y. Ebara, et al. (2023)

- New test site was constructed.
- Transportation and assembly of SC230 were completed in 2020.
- Commissioning was started at the end of 2020.



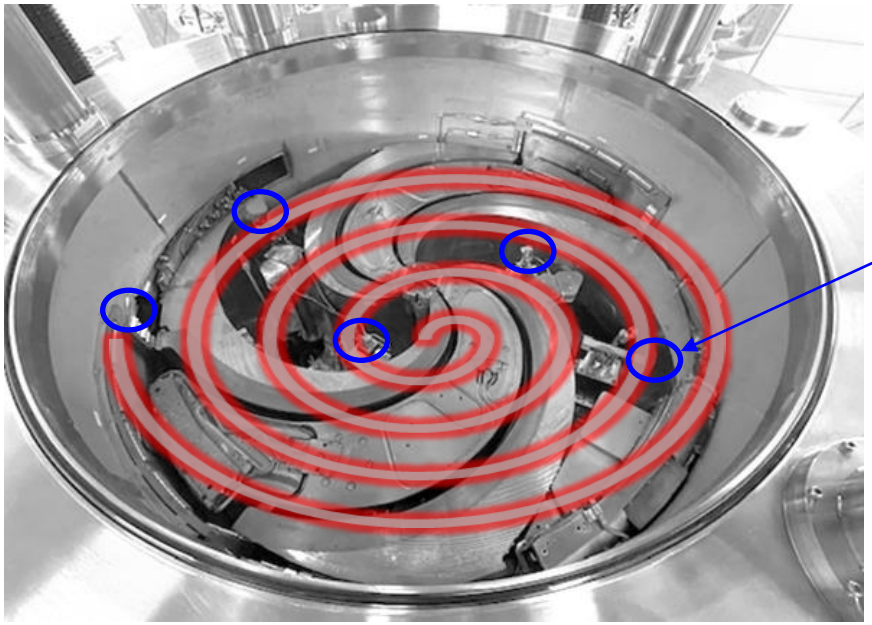


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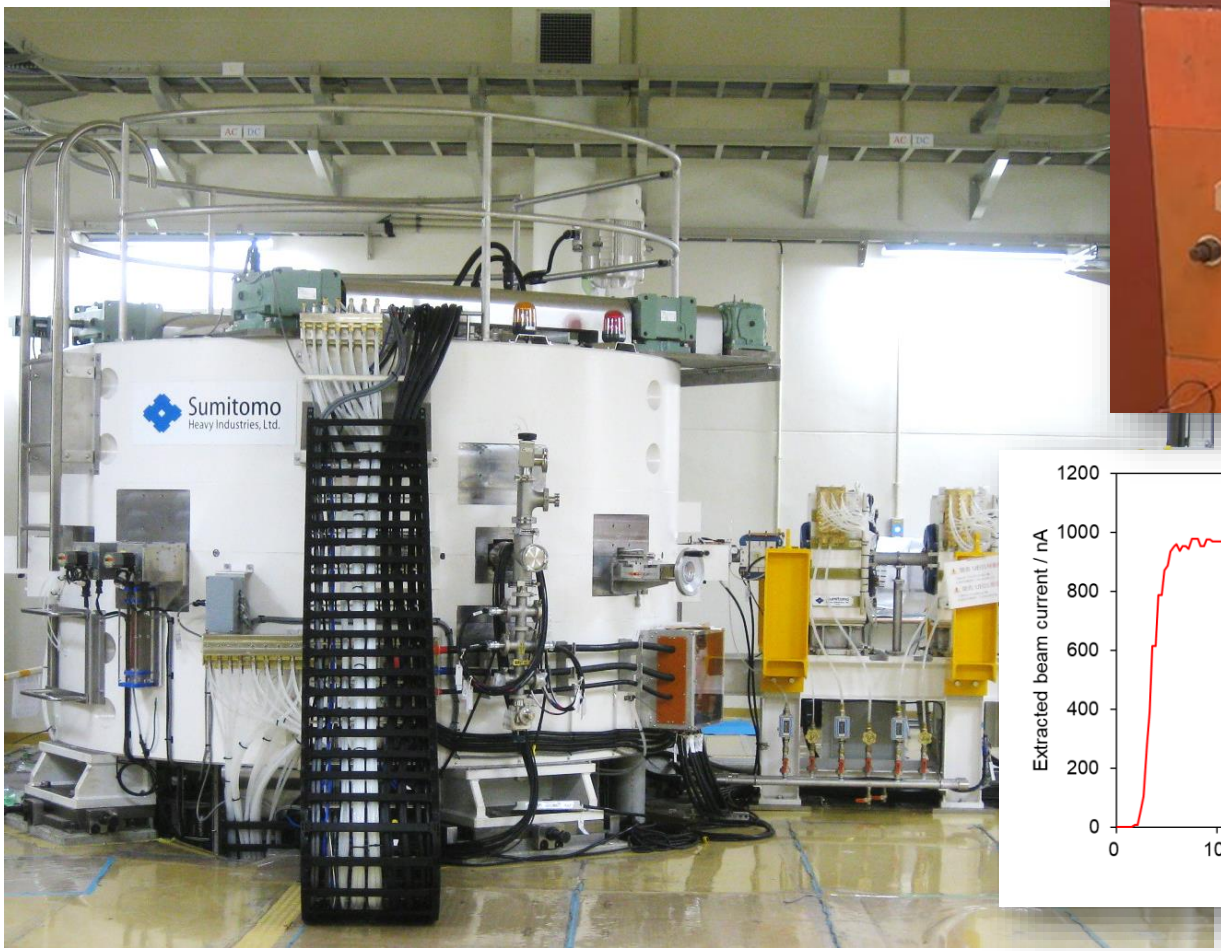
During commissioning, probes installed at each radius were used to check beam conditions, and parameters were adjusted.

Commissioning : First beam observation

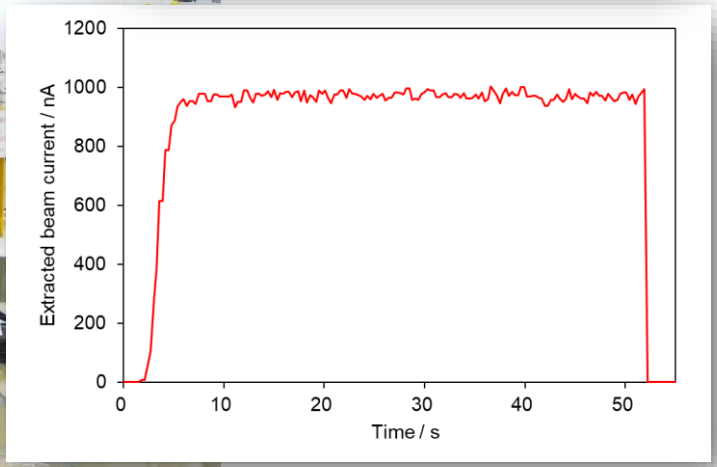
Sumitomo Heavy Industries, Ltd.

Y. Ebara, et al. (2023)

- **First beam was observed in Jul. 2021.**





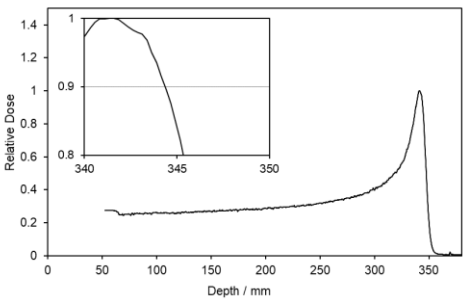


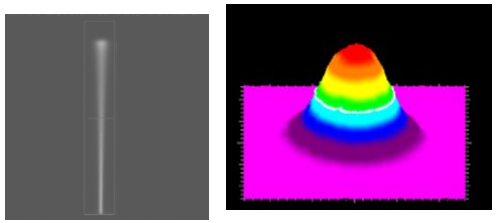
Commissioning : Beam status

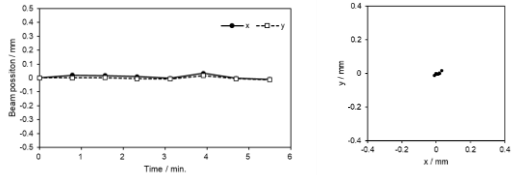
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Parameters	Specifications	Result	Pass/Fail
Max. beam current	1000 nA	>1000 nA	✓ Pass
Beam energy	233-238 MeV	237 MeV	✓ Pass
Energy repeatability	< 0.2 MeV	0.18 MeV	✓ Pass
Extraction efficiency	≥ 60%	67%	✓ Pass
Min. controllable beam current	< 1 nA	< 1nA	✓ Pass
Ripple current	≤ 2 % (1σ)	0.6%	✓ Pass
Beam current stability	≤ 1 % (1σ) for 2 min.	1.0%	✓ Pass
Beam responsivity	≤ 50 μsec	~30 us	✓ Pass
Beam position stability	< ±0.1 mm for 2 min.	< ±0.05 mm	✓ Pass
RMS emittance	< 2.2 μm (x) < 1.4 μm (y)	1.4 μm (x) 0.7 μm (y)	✓ Pass
Total power consumption in operation	< 200 kW	193 kW	✓ Pass

Y. Ebara, et al. (2023)





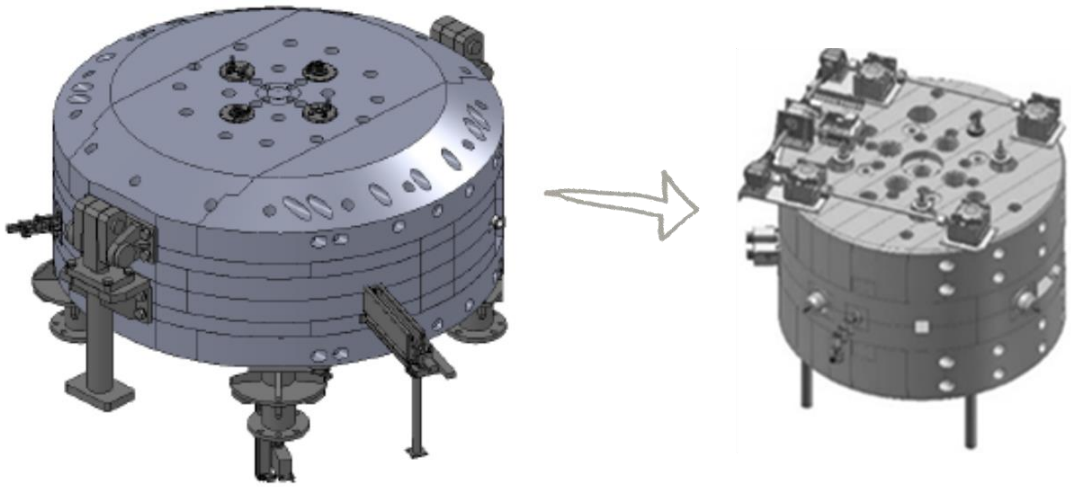


A beam that satisfies all required specifications was obtained.

Commissioning : Result

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- **Commissioning completed in Nov. 2021.**

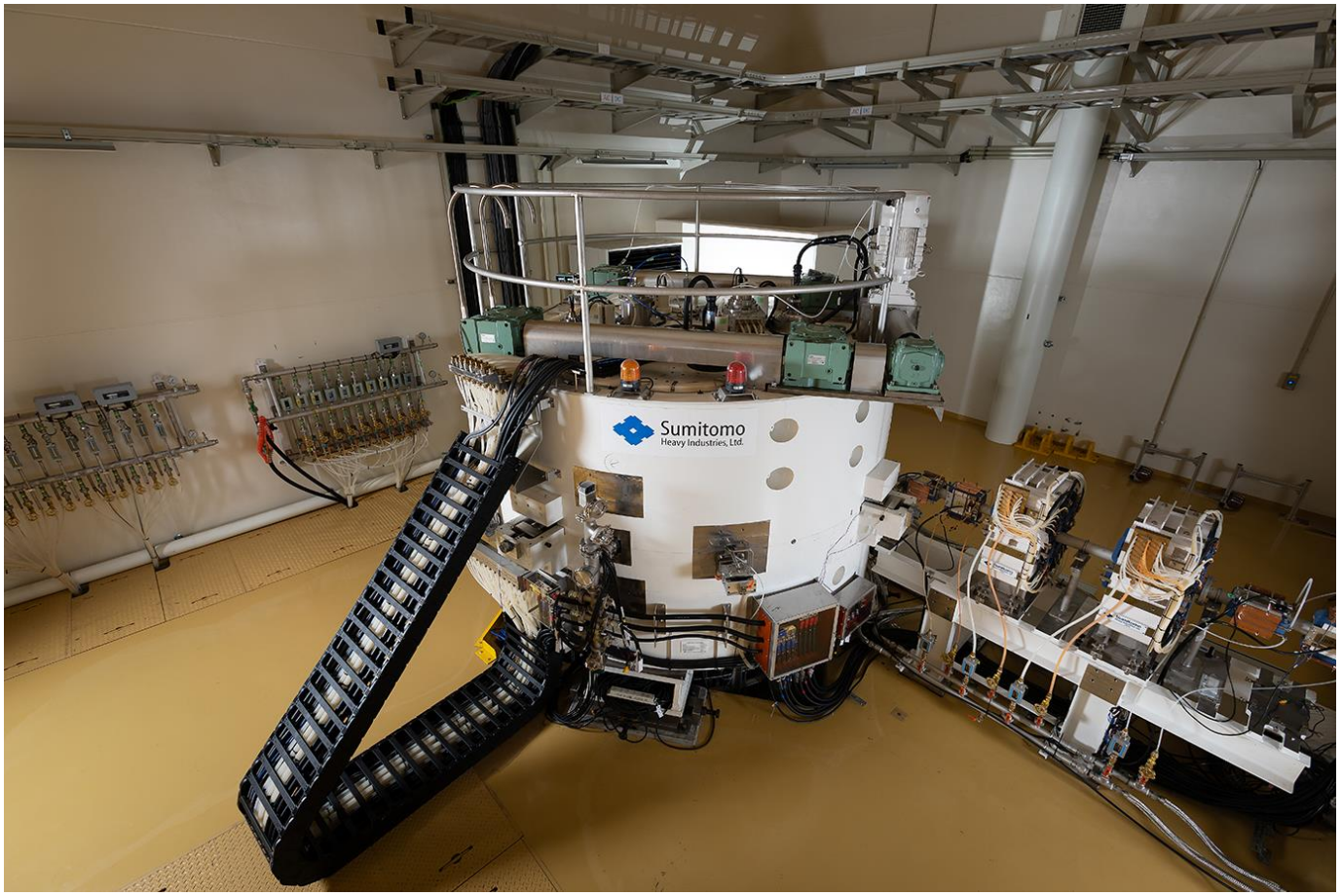


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Commissioning : Result

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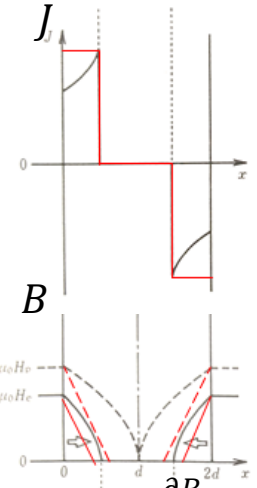
- **Commissioning completed in Nov. 2021.**



Effect of SCIF on beam

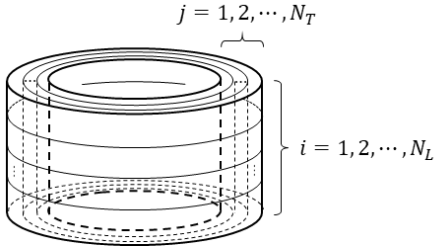
- Calculation of SCIF

item	assumptions
Current density distribution	Critical current model (Bean-London model)
Superconducting properties	Scaling-low
Jc distribution within the filament	Ellipse approximation model
Interaction between filaments	Ignore
Magnetic field distribution inside the filament	Uniform
Magnetization distribution within the divided area	Uniform



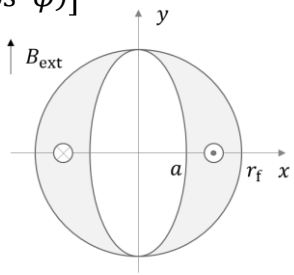
$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J_c} \Rightarrow \frac{\partial B_y}{\partial x} = \mu_0 J_c$$

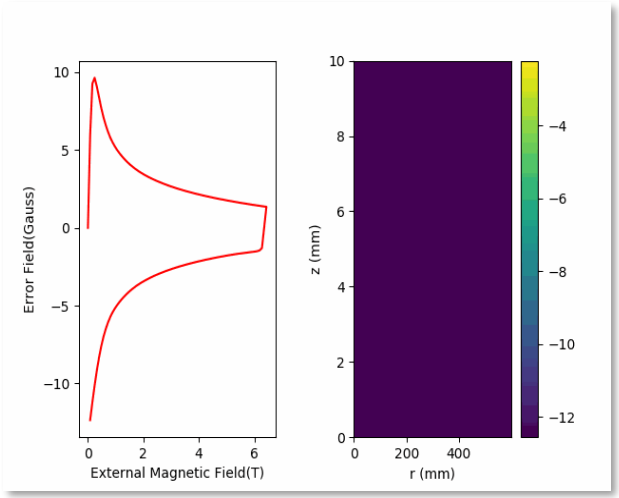
$$B_{\mathrm{SCIF}}(P) = \sum_{i=1}^{N_L} \sum_{j=1}^{N_T} \left[\int_0^{2\pi} d\theta \cdot b_{z_{ij}}(\theta) \right]$$



$$b_z(\theta) = \frac{\mu_0}{4\pi l'^3} \left[3m_{r'} \sin\phi \cos\phi + m_z \left(2\sin^2\phi - \cos^2\phi \right) \right]$$

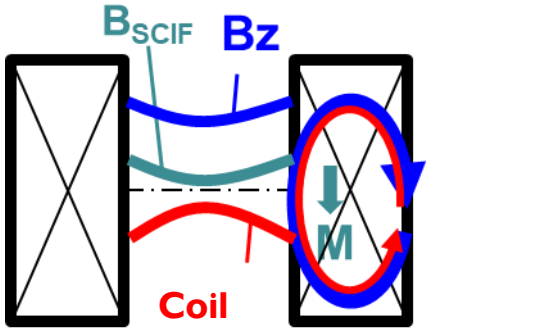
$$\boldsymbol{m} = A\boldsymbol{M} = \frac{4}{3\pi} A J_c r_f \left[1 - \left(\frac{a}{r_f} \right)^2 \right] \cdot \frac{\boldsymbol{B}}{|B|}$$

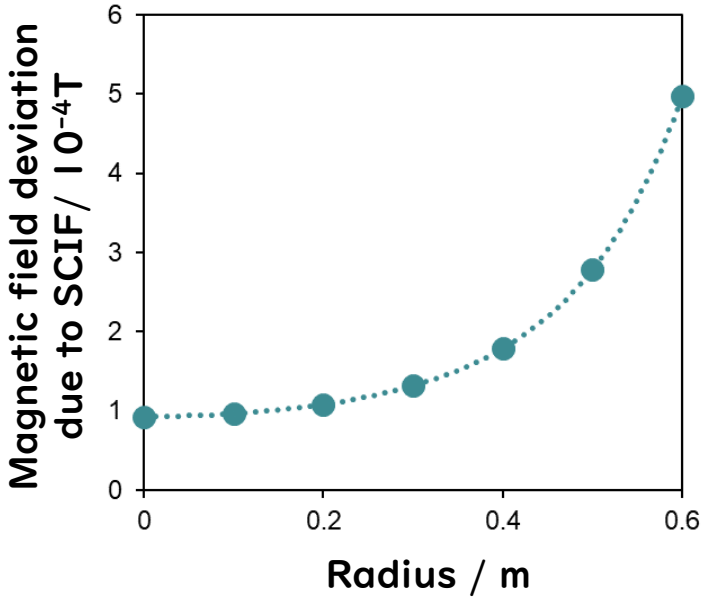


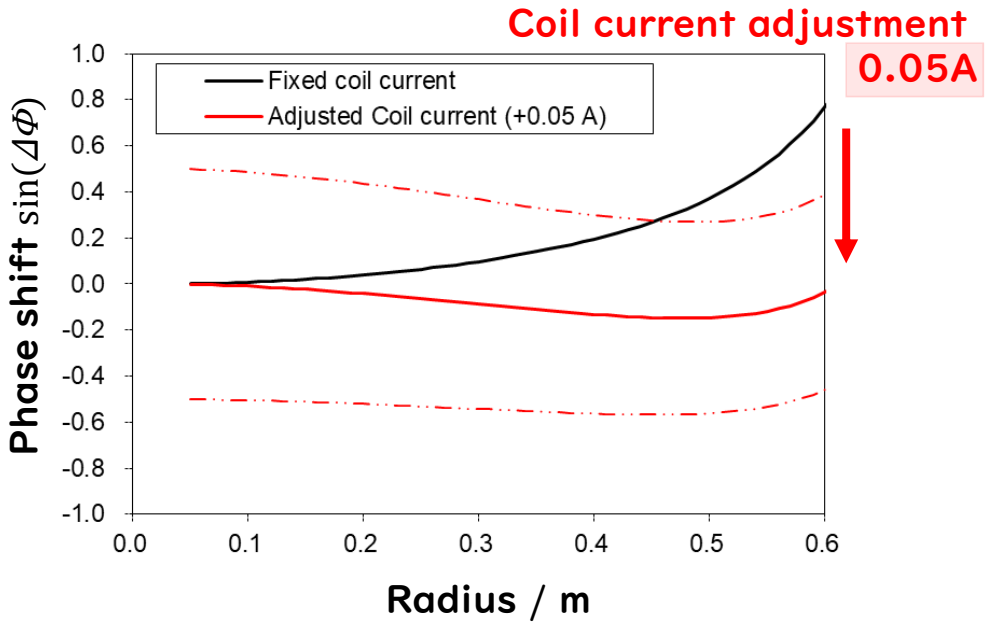


Effect of SCIF on beam

- SCIF and phase shift in SC230

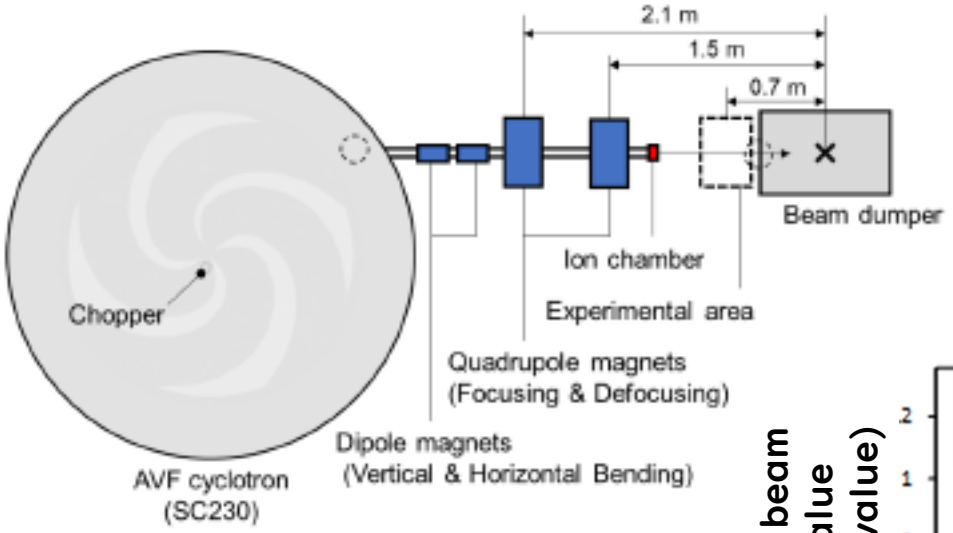


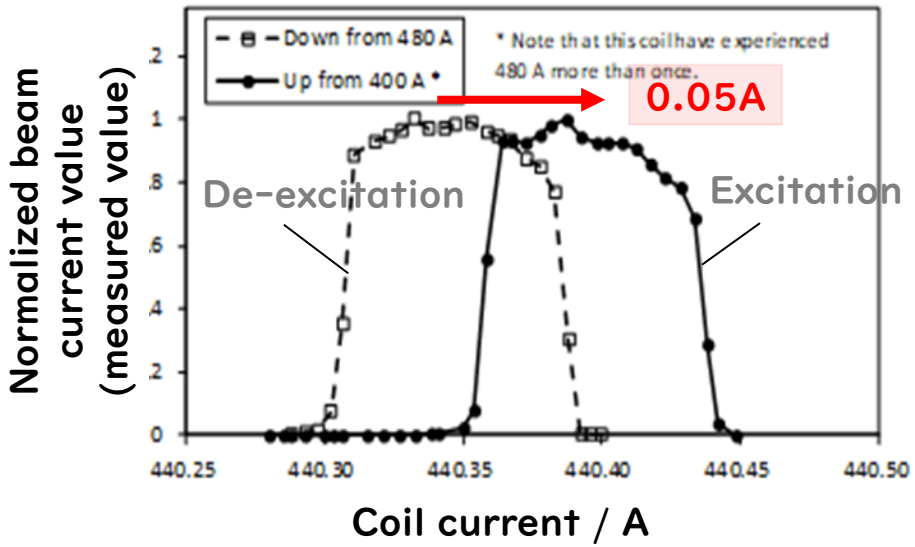




Effect of SCIF on beam

- Dependence of beam current on coil current history



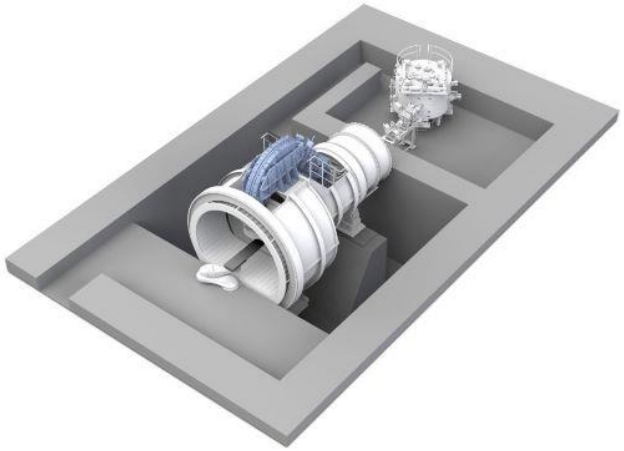


Conclusion

- SC230 is a superconducting AVF cyclotron, which features the compact size weighing 65 tons, high current beam of 1μA, low energy consumption less than 200 kW.
- It was developed and the commissioning was completed in 2021.



Next-generation proton beam therapy system using SC230



SC AVF cyclotron SC230 for PT has been developed.

First order from Taichung Veteran's General Hospital

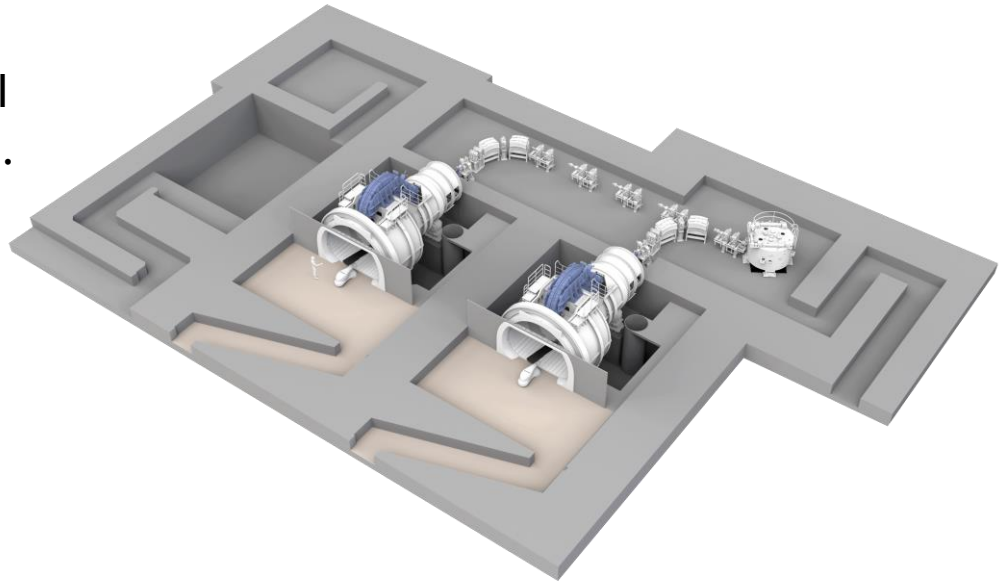
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台中栄民総医院
Taichung Veterans General Hospital

Located at the foot of Mt. Dadu in Taichung, it is the only public medical center in central Taiwan, with a total of about 4,250 employees, 1,624 beds, and a total of 7,700 outpatient visits daily.



- We have received an order from Taichung Veterans General Hospital for CyBeam proton therapy system.
- Construction of the new building has begun, and the equipment is expected to be delivered in 2025.



SC230 will contribute to PT soon.

Thank you

Papers related to this presentation

Sumitomo Heavy Industries, Ltd.

Reference	Content	QR code
H. Tsutsui, et al. "Design study of a superconducting AVF cyclotron for proton therapy." Proc. 20th Int. Conf. on Cyclotrons and their Applications (Cyclotrons13). (2013).	Basic design	
J. Yoshida, et al. "Excitation test of superconducting magnet in 230-MeV isochronous cyclotron for proton therapy." IEEE Transactions on Applied Superconductivity 30.4 (2019): 1-5.	Test of magnet	
Y. Ebara, et al. "Field mapping system and field adjustment for a 230-MeV proton cyclotron." Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment 953 (2020): 163186.	Mapping (Magnetic field measurement)	
Y. Ebara, et al. "Performance of cryogen-free superconducting magnet in isochronous cyclotron for proton therapy." IEEE Transactions on Applied Superconductivity 33.5 (2023): 1-4.	Performance of magnet	
Y. Ebara, et al. "Commissioning of the Sumitomo superconducting AVF cyclotron SC230." Proceedings of the 23rd Int. Conf. on Cyclotrons and their Applications. (2023).	Commissioning	